Jiaqi Ge

ge24876@gtiit.edu.cn

EDUCATION

Guangdong Technion – Israel Institute of Technology Sep 2023 – Present

B.Sc. in Mathematics with Computer Science | English-taught Program | GPA: 89.7/100

PUBLICATIONS

Manuscript on Repository Knowledge Acquisition for Software Issue Resolution

Co-author | Under submission

RESEARCH EXPERIENCE

Research Assistant — Repository QA for Software Issue Resolution Feb 2026 – Present

Shanghai Jiao Tong University | advised by Prof. Xiaodong Gu

- Contributed to manuscript writing and revision, including related work, experimental results, method analysis, ablation discussion, and figure/result interpretation.
- Designed and executed baseline vs. QA-augmented experiments for repository-level software issue resolution.
- Built a judge-based evaluation pipeline for question-level, set-level, and pairwise assessment, and verified the scoring and aggregation process.
- Conducted fail-to-pass trajectory analysis, including repair-stage segmentation, QA-usage analysis, and patch-alignment analysis, and prepared supporting figures and reports.

Research Assistant — Mixture-of-Experts (MoE) Apr 10, 2026 – Present

MBZUAI – Mohamed bin Zayed University of Artificial Intelligence | advised by Prof. Lijie Hu

- Working on a research project related to Mixture-of-Experts (MoE).

Research Assistant — Explaining AI Models through Metamorphic Property Generation Oct 2025 – Feb 2026

Guangdong Technion – Israel Institute of Technology | advised by Prof. Nazareno Aguirre

- Built a grammar-based fuzzing pipeline to generate multi-step text perturbation chains (typos, punctuation, entity swaps, negation, …) using Checklist.
- Implemented candidate metamorphic-property generation and checking (INV invariance / DIR directional) to evaluate model behavior under k-step transformations.
- Integrated a HuggingFace sentiment-analysis model with Checklist to run automated INV/DIR test suites at scale, and summarized experimental findings for research documentation.

Research Assistant — Numerical Methods for Boundary Value Problems Aug 2025 – Nov 2025

Guangdong Technion – Israel Institute of Technology | advised by Prof. Antti Rasila | Project Score: 95/100

- Conducted research on numerical solutions for boundary value problems (BVPs) in MATLAB.
- Reviewed classical theory including existence/uniqueness, Green's functions, and Sturm–Liouville problems.
- Compared shooting, finite-difference, and spectral methods for linear and nonlinear BVPs.
- Used the MATLAB Chebfun toolbox to solve nonlinear systems, singular Sturm–Liouville problems, and chaotic dynamical systems.
- Authored a 20-page report titled "Solving Boundary Value Problems Using Chebfun".

Research Assistant — Integration of Multi-Age Clocks for Alzheimer's Disease Diagnosis May 2024 – Jun 2025

University of Nebraska Medical Center (UNMC) | advised by Dr. Shibiao Wan

- Integrated brain-aging and epigenetic aging models to support Alzheimer's disease (AD) diagnosis, and analyzed relationships among different age clocks for age-related disease prediction.
- Contributed to the implementation and code review of a multi-modal age-clock integration framework combining DNA methylation and neuroimaging data.

- Conducted literature reviews on biological aging biomarkers and age estimation models.

TEACHING EXPERIENCE

Undergraduate Teaching Assistant — Linear Algebra & Calculus Mar 2026 – Present

Guangdong Technion – Israel Institute of Technology | *Shantou, China*

- Facilitate student learning in weekly workshops by providing guidance on complex mathematical problems and theorems.
- Assist students in mastering foundational concepts of Linear Algebra and Calculus through interactive problem-solving sessions.

Undergraduate Teaching Assistant — Python Oct 2025 – Jan 2026

Guangdong Technion – Israel Institute of Technology | *Shantou, China*

- Assisted in teaching the introductory Python course and supervised lab sessions.
- Supported students with debugging code and understanding core programming concepts.

AWARDS

Scholarship — 28,500 USD 2023

Awarded by Guangdong Technion – Israel Institute of Technology for performance in the Chinese National College Entrance Examination (Gaokao).

SKILLS

Programming: Python, C, C++, Java, MATLAB, Assembly
Languages: Chinese (Native), English (Fluent)